

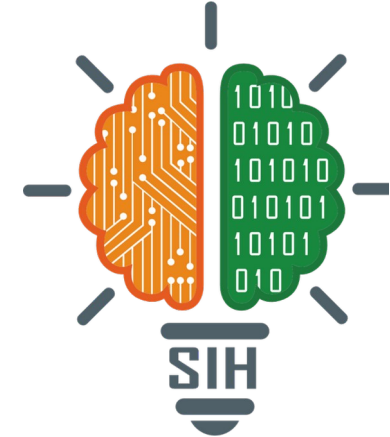
SMART INDIA HACKATHON 2026



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2026

TITLE PAGE

- **Problem Statement ID** - *<fill from SIH portal>*
- **Problem Statement Title**- Real-time Airfare Price Index for the CPI Transport & Communication sub-group
- **Theme**- *<fill from SIH portal>*
- **PS Category**- Software
- **Team ID**- *<fill from SIH portal>*
- **Team Name (Registered on portal)**- *<fill from SIH portal>*



WHAT THE PROTOTYPE ALREADY DOES, EVERY NIGHT



A block, a 403 or a CAPTCHA is recorded as non-response and imputed — never evaded. Every point published with its coverage, imputation share and quality flag.

APIx

A compliant, daily-frequency airfare price index — collected only where robots.txt permits, chained from geometric-mean cell prices, and published with its own coverage and quality on every point.

975

basket cells

100.0

base 01-Apr-26

74

tests passing

IDEA TITLE

APIx — Real-time Airfare Price Index



Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- Collect public domestic fares daily from airline and OTA sites — only where robots.txt permits — rendering JS pages in a real Chromium browser at a 5-second crawl delay.
- Normalise every quote to the all-in fare a household pays: base + GST + UDF/PSF/ASF + mandatory surcharges. Optional seats, bags and meals are excluded.
- The item is a cell = source × route × carrier × advance-purchase window × cabin — a basket of 975 cells (15 city pairs × 6 carriers × 5 booking windows × Economy).
- Cells collapse to one geometric-mean price a day and chain into a weighted index anchored at 100.0 on 01-Apr-2026, published as daily / weekly / monthly series with a JSON API.

How it addresses the problem

- CPI air-fare collection is monthly and manual, while a fare can change several times a day — APIx makes that item observable daily, automatically.
- Every published point carries coverage, imputation share and a quality flag, so a reader sees how much of the basket was really observed.
- The raw quote archive is append-only, so the whole series is recomputable and independently replicable.

Innovation and uniqueness of the solution

- The source sits inside the item identity — when a site goes dark, coverage falls but no fake price movement enters the index.
- Ethically designed for real: a 403, a CAPTCHA or a bot challenge is recorded as non-response and never evaded — a unit test fails the build if evasion machinery is added. That is what keeps the number admissible as official statistics.
- Our own RFC 9309 robots.txt parser: Python's stdlib parser ignores wildcards and fails permissive — it would have claimed permission we do not have.
- The pipeline refuses to publish a point it cannot stand behind (exit code 4) instead of quietly freezing the site behind a green build.

PROTOTYPE STATUS

Working end to end

975

cells in the basket

3 of 9

sources permit collection

74

tests, ~12 s

0.05%

endpoint error vs. known truth

Verified on a 90-day stream with known drift, a +12% shock and 6% non-response.

TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

COLLECTION & INDEX ENGINE

Python 3.11 · Playwright /
Chromium · PyYAML · pytest

DATA

PostgreSQL (Neon) · psycopg 3 ·
SQL migrations

WEB

Next.js 16 · React 19 · TypeScript
strict · Tailwind 4

RUNTIME

GitHub Actions 02:30 UTC · Vercel ·
FastAPI (local)

Methodology and process for implementation

1 Basket

975 cells from
config/basket.yaml

2 robots.txt gate

RFC 9309 parser,
fails closed

3 Fetch

Chromium renders JS,
5 s crawl delay

4 Parse

one adapter per site;
a block = non-response

5 raw_quote

append-only archive
in Postgres

6 Normalise

all-in fare, currency,
fare family

7 Quality control

range checks + MAD
outliers on log fares

8 Cell price

geometric mean per
cell per day

9 Impute & chain

class → route → all-items;
 $I(t) = I(t-1) \times R(t)$

10 Publish

index_point → dashboard,
/api/index, /api/inflation

Chained weighted geometric (Young) index

$$R(t) = \exp\left(\sum_c \tilde{w}_c \cdot \ln\left[p(c,t) / p(c,t-1)\right]\right) \quad I(t) = I(t-1) \times R(t)$$

c runs over cells seen in both periods; $\tilde{w}$ is the basket weight renormalised over that matched set. The monthly figure is a direct month-over-month comparison, not a product of 30 daily links.

Nightly job — every step can stop the run

tests → robots audit → collect → rebuild → report

A failing test, a newly disallowed source, an empty pass or an unpublishable point each exits non-zero — the report still runs.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

✓ Already built

The prototype runs end to end today: collection, index engine, database, dashboard and API.

✓ Cheap to run

One nightly GitHub Actions job plus a read-only Vercel site on managed Postgres — free tiers.

✓ Verified maths

74 tests, index paths worked out by hand; 0.05% endpoint error against a known synthetic truth.

✓ Extensible

A new site — or a partner data feed — drops in as one adapter file; the index engine is untouched.

Potential challenges and risks

Coverage is sparse and uneven

Only 3 of 9 audited sites allow the fare paths an index needs. IndiGo, SpiceJet and Cleartrip disallow them; Akasa and MakeMyTrip are unreadable, so we fail closed.

Non-response can masquerade as inflation

It already did: 28–30 Aug 2026, two sources returned nothing, imputation hit 84.9%, and the index drifted +8.2% in three days while the build stayed green and the site stayed frozen.

One slow robots.txt read cost a whole source

A single timeout poisoned the cache for an hour, turning all 75 of that source's requests into non-responses.

Weights are placeholders, not statistics

Basket weights are shaped like DGCA figures but are not them, and O-D and booking-lead shares are combined assuming independence.

Strategies for overcoming these challenges

Source is part of the cell and weights renormalise over matched cells, so a silent source costs coverage, never a fake price move. The real fix is data access, not better scraping: a partner feed, GDS data or reporting under the Collection of Statistics Act arrives as one more adapter.

Donor hierarchy imputes from the cell's own stratum (class → route → all-items); quality flags warn at 35% and fail at 60%; the rebuild now exits 4 rather than publishing a fail-quality point, naming the coverage, the bad days and which sources went silent.

The gate now separates a refusal from a network failure: an HTTP status is the operator answering — one attempt, cached an hour; a timeout is the network — three attempts, cached 120 seconds.

Flagged in basket.yaml, METHODOLOGY.md and the README's honest-status section; replace with DGCA monthly O-D traffic and GDS booking-lead data, on an annual basket review, before anything is published as a statistic.

IMPACT AND BENEFITS

Potential impact on the target audience

MoSPI / NSO

CPI compilers of the Transport & Communication sub-group

RBI & analysts

A fast-moving component read daily instead of once a month

MoCA / DGCA

Fare behaviour by route, carrier and booking window

Researchers

An append-only archive that lets anyone replicate the series

Travellers & media

A public advance-purchase curve: how far ahead fares actually change

Daily

instead of monthly observation of the air-fare item

Every point

ships coverage, imputation share and a quality flag

Zero

access controls circumvented — so the number is admissible

Reproducible

the whole index recomputes from the raw quote archive

Benefits of the solution (social, economic, environmental, etc.)

ECONOMIC

- An earlier, higher-frequency signal on transport inflation, where the current one lags by weeks.
- Base fare and tax are kept per quote, so an ex-tax variant can sit beside the headline — separating a GST change from a genuine fare move.
- Automated collection removes recurring manual price-collection effort for this item.

SOCIAL

- Open methodology and a public compliance page: which sites were asked, what they answered, and why.
- Households can see the advance-purchase curve and the route x window fare matrix rather than guessing when to book.
- Any series built on simulated data is banner-flagged, with no switch to turn the warning off.

GOVERNANCE & ENVIRONMENTAL

- Statistics produced without defeating any site's access controls — defensible in a way scraped-by-evasion data is not.
- The audit trail, the collection log and the raw archive make independent replication possible.
- One nightly job and a read-only site: a negligible compute footprint and no field travel to collect prices.

Details / Links of the reference and research work

PROJECT ARTEFACTS

1 Source code, docs and issue tracker

github.com/Prasun-Acharjee/apix-airfare-index

2 METHODOLOGY.md

the formal spec: price concept, the cell, the geometric mean, chaining, the imputation hierarchy and the quality flags

3 DEPLOY.md

Vercel + Neon and self-hosted Docker deployment

4 Live pages

[/ index chart and route x window matrix](#) · [/compliance robots audit and collection log](#) · [/methodology](#)

5 Public API

[/api/index/{daily|weekly|monthly}](#) · [/api/inflation](#) · [/api/routes](#) · [/api/compliance](#) · [/api/collection-log](#)

6 Test suite — 74 tests

including `test_stdlib_parser_would_have_been_wrong` and `test_codebase_contains_no_evasion_machinery`, which pin the two claims this slide deck makes

STANDARDS, SOURCES AND PRIOR ART

1 RFC 9309 — Robots Exclusion Protocol

rfc-editor.org/rfc/rfc9309.html — implemented in `apix/compliance/rfc9309.py`, because Python's `urllib.robotparser` handles neither wildcards nor full-URL directives and fails permissive

2 CPI Manual: Concepts and Methods — ILO, IMF, OECD, UNECE, Eurostat

chained indices, elementary aggregates and the imputation of missing prices

3 Eurostat and ONS practice on web-scraped prices

treatment of airfares in the HICP/CPI and the use of scraped data in official statistics

4 MoSPI — Consumer Price Index

the Transport & Communication sub-group this index is scoped to

5 DGCA — Monthly Traffic Statistics

origin-destination passenger traffic and carrier market share, the intended replacement for the placeholder basket weights

6 Collection of Statistics Act, 2008

the statutory route to airline data access that removes the coverage problem entirely

7 Playwright · Next.js · PostgreSQL documentation

JS rendering, the App Router read path and the schema